



Vaxcel® CEFUROXIME INJECTION

COMPOSITION:

Each vial contains Cefuroxime Sodium equivalent to Cefuroxime 750mg and 1.5g.

PRESENTATION:

White or almost white faintly yellow powder.

INDICATIONS:

Respiratory tract infections :

Acute and chronic bronchitis, infected bronchiectasis bacterial pneumonia, lung abscess and post-operative chest infection.

Ear, nose and throat infections:

Sinusitis, tonsillitis and pharyngitis.

Urinary tract infections:

Acute and chronic pyelonephritis, cystitis and asymptomatic bacteriuria.

Soft-tissue infections:

Cellulitis, erysipelas, peritonitis and wound infections.

Bone and joint infections:

Osteomyelitis and septic arthritis.

Obstetric and gynaecological infections:

Pelvic inflammatory diseases.

Gonorrhoea:

Particularly when penicillin is unsuitable.

Other infection:

Septicaemia and meningitis.

Prophylaxis:

Against infection in abdominal, pelvic, orthopaedic, cardiac, pulmonary, oesophageal and vascular surgery where there is increased risk from infection.

PHARMACOLOGY:

Mechanism of Action:

Cefuroxime is a bactericidal cephalosporin antibiotic which is resistant to most β -lactamases and is active against a wide range of Gram-positive and Gram-negative organism. The bactericidal action of cefuroxime results from inhibition of cell-wall synthesis. It is indicated for the treatment of infections before infecting organism has been identified or when caused by sensitive bacteria. In addition it is an effective prophylactic against post-operative infection in a variety of operations. It may be

used in combination with an aminoglycoside antibiotic, or in conjunction with metronidazole (orally or by suppository or injection), especially for prophylaxis in colonic surgery. Cefuroxime is highly active against *Staphylococcus aureus*, including strains which are resistant to penicillin (but not the rare methicillin resistant strains), *Staphylococcus epidermis*, *Haemophilus influenzae*, *Klebsiella spp.*, *Enterobacter spp.*, *Streptococcus pyogenes*, *Escherichia coli*, *Str. Mitis*, *Clostridium spp.*, *Proteus mirabilis*, *P. Rettgeri*, *Salmonella typhi*, *S. typhimurium* and other *Salmonella spp.*, *Shigella spp.*, *Neisseria spp.* and *Bordetella pertussis*. It is also moderately active against strains of *Pr. vulgaris*, *Pr. morganii* and *Bacteroides fragilis*. *In vitro* the activities of Cefuroxime and aminoglycoside antibiotics in combination have been shown to be at least additive with occasional evidence of synergy.

Pharmacokinetics:

Peak levels of Cefuroxime are achieved within 30 to 45 minutes after intramuscular administration. The serum half-life after either intramuscular or intravenous injection is approximately 70 minutes. Concurrent administration of probenecid prolongs the excretion of the antibiotic and produces an elevated peak serum level. There is almost complete recovery of unchanged Cefuroxime in the urine within 24 hours of administration, the major part being eliminated in the first six hours. Approximately 50% is excreted through the renal tubules and approximately 50% by glomerular filtration. Concentrations of Cefuroxime in excess of the minimum inhibitory levels for common pathogens can be achieved in bone, synovial fluid and aqueous humor. Cefuroxime passes the blood-brain barrier when the meninges are inflamed. Cefuroxime is approximately 50% bound to serum protein.

DOSAGE & ADMINISTRATION:

Adults:

Many infections will respond to 750mg tds by IM or IV injection. For more severe infections, this dose should be increased to 1.5g tds IV. The frequency of IV injections can be increased to six hourly if necessary, giving total doses of 3g to 6g daily.

Infants and Children:

Doses of 30 to 100mg/kg/day IV in 3 or 4 divided doses. A dose of 60mg/kg/day will be appropriate for most infections.

Neonates:

Doses of 30 to 100mg/kg/day given as 2 or 3 divided doses. In the first few weeks of life the serum half-life of Cefuroxime can be 3-5 times that in adults.

Other Recommendations:

Gonorrhoea: 1.5g should be given as a single dose. This may be given as 2 x 750mg injections into different sites, eg each buttock. Meningitis: Cefuroxime is suitable for sole therapy of bacterial meningitis due to sensitive strains. The following dosages are recommended: Infants and children: 200 - 240mg/kg/day IV in 3 or 4 divided doses. This dosage may be reduced to 100mg/kg/day IV after 3 days or when clinical improvement occurs. Neonates: The initial dosage should be 100mg/kg/day IV. A reduction to 50mg/kg/day IV may be made when clinically indicated. Adults: 3g IV every 8 hours. Data are not yet sufficient to recommend a dose for intrathecal administration.

Prophylaxis: The usual dose is 1.5g IV with induction of anaesthesia for abdominal, pelvic and orthopaedic operations, but may be supplemented with two 750mg IM doses 8 and 16 hours later. In cardiac, pulmonary, oesophageal and vascular operations, the usual dose is 1.5g IV with induction of anaesthesia continuing with 750mg IM 3 times a day for a further 24 - 48 hours. In total joint replacement, 1.5g cefuroxime powder may be mixed dry with each pack of methyl methacrylate cement monomer before adding the liquid polymer.

Dosage In Impaired Renal Function:

Cefuroxime is excreted by the kidneys. Therefore, as with all such antibiotics, in patients with impaired renal function it is recommended that the dosage of Cefuroxime should be reduced to compensate for its slower excretion. However, it is not necessary to reduce the dose until the creatinine clearance falls below 20ml/min. In adults with marked impairment (creatinine clearance 10-20ml/min) 750mg twice daily is recommended and with severe impairment (creatinine clearance <10-20ml/min) 750mg daily is adequate. For patients on dialysis a further 750mg dose should be given at the end of each dialysis. When continuous peritoneal dialysis is being used, a suitable dosage is usually 750mg twice daily.

Males:

$$\text{Creatinine Clearance (ml/min)} = \frac{\text{Weight(kg)} \times (140) - (\text{age in years})}{72 \times \text{serum creatinine (mg/dl)}}$$

Female:

0.85 x male value

Compatibility And Stability:

Protect from light. Cefuroxime should not be mixed in the syringe with aminoglycoside antibiotics. Suspension of Cefuroxime for intramuscular injection and aqueous solutions for direct intravenous injection retain their potency for five hours if kept below 25°C and for 48 hours if refrigerated. More dilute solutions, eg. 1.5g plus 50ml water for injections, retain satisfactory potency for 24 hours if kept below 25°C and for 48 hours if refrigerated. Some increase in the color of prepared solutions and suspensions of Cefuroxime may occur on storage. 1.5g Cefuroxime constituted with 15ml water for injection may be added to metronidazole injection (500mg/100ml) and both retain their activity for up to 24 hours below 25°C. 1.5g Cefuroxime is compatible with azlocillin 1g (in 15ml) or 5g (in 50ml) for up to 24 hours at 4°C or 6 hours below 25°C. Cefuroxime (5mg/ml) in 5% w/v or 10%w/v xylitol injection may be stored for up to 24 hours at 25°C. Cefuroxime is compatible with the more commonly used intravenous infusion fluids. It will retain potency for up to 24 hours at room temperature in Sodium Chloride Injection BP 0.9%w/w, 5% Dextrose Injection BP, 0.18%w/v Sodium Chloride Plus, 4% Dextrose Injection BP and Compound Sodium Lactate Injection BP (Hartmann's Solution). The pH of 2.74%w/v Sodium Bicarbonate Injection BP considerably affects the colour of the solution and therefore this solution is not recommended for the dilution of Cefuroxime. However, if required, for patients receiving Sodium Bicarbonate Injection by infusion the Cefuroxime may be introduced into the tube of the giving set. The stability of Cefuroxime in Sodium Chloride Injection BP 0.9%w/v and 5% Dextrose Injections is not affected by the presence of Hydrocortisone Sodium Phosphate. Cefuroxime is also compatible with aqueous solutions containing up to 1% Lignocaine Hydrochloride.

Administration Intramuscular :

Add 1ml water for injections to 250mg Cefuroxime or 3ml water for injections to 750mg Cefuroxime. Shake gently to produce an opaque suspension.

Administration Intravenous:

Dissolved Cefuroxime in water for injections using at least 2ml for 250mg, at least 6ml for 750mg, or 15ml for 1.5g. For short intravenous infusion (eg up to 30 minutes), 1.5g may be dissolved in 50ml water for injections. These solutions may be given directly into the vein or introduced into the tubing of the giving set if the patient is receiving parenteral fluids. Cefuroxime is compatible with the more commonly used intravenous fluids.

CONTRAINDICATIONS:

Patients known to be hypersensitive to Cephalosporin antibiotics.

WARNING & PRECAUTIONS:

Cephalosporin antibiotics may in general be given safely to patients who are hypersensitive to penicillin, although cross-reactions have been reported. Special care is indicated in patients who have experienced an anaphylactic reaction to penicillin. Cephalosporin antibiotics at high dosage should be given with caution to patients receiving concurrent treatment with potent diuretics such as frusemide, as these combinations are suspected of adversely affecting renal function. Clinical experience with Cefuroxime has shown that this is not likely to be a problem at the recommended dose levels. There is no experimental evidence of embryopathy or teratogenic effects attributable to Cefuroxime. Before therapy with Cefuroxime is instituted, carefully inquiry should be made to determine whether the patients has had previous hypersensitivity reactions to cephalosporins, penicillins, or other drugs. This product should be given cautiously to penicillin-sensitive patients. Antibiotics should be administered with caution to any patient who has demonstrated some form of allergy, particularly to drugs. If an allergic reaction to Cefuroxime occurs, discontinue the drug. Serious acute hypersensitivity reactions may require epinephrine and other emergency measure. Pseudomembranous colitis has been reported with nearly all antibacterial agents, including cefuroxime, and may range in severity from mild to life threatening. Therefore, it is important to consider this diagnosis in patients who present with diarrhea subsequent to the administration of antibacterial agents. Serious and occasionally fatal hypersensitivity reactions (including anaphylactoid and severe cutaneous adverse reactions) have been reported in patients receiving therapy with beta-lactams. Before initiating therapy with Vaxcel Cefuroxime Injection, careful inquiry should be made concerning previous hypersensitivity reactions to penicillins, cephalosporins, carbanems or other beta-lactam agents. If an allergic reaction occurs, Vaxcel Cefuroxime Injection must be discontinued immediately and appropriate alternative therapy instituted.

Use in Pregnancy and Lactation:

It should be administered with caution during the early months of pregnancy.

SIDE EFFECTS :

Adverse reactions have occurred relatively infrequently and have been generally mild and transient in nature. Effects

reported include rashes and gastrointestinal disturbance. Prolonged use with other antibiotics may result in the overgrowth of non-susceptible organisms, eg. Candida. The principle changes in haematological parameters seen in some patients have been of decreased haemoglobin concentration and of eosinophilia. A positive Coombs' test has been found in some patients treated with cefuroxime, this phenomenon can interfere with the cross-matching of blood. Although there are sometimes transient rises in serum liver enzymes or serum bilirubin, particularly in patients with pre-existing liver disease, there is no evidence of hepatic involvement. There may be some variation in the results of biochemical tests of renal function, but these do not appear to be of clinical importance. As a precaution, renal function should be monitored if this is already impaired. Transient pain may be experienced at the site of intramuscular injection as this is more likely to occur with higher doses. However, it is unlikely to be a cause of discontinuation of treatment. Occasionally, thrombophlebitis may follow IV injection.

DRUG INTERACTIONS:

Concomitant oral administration of probenecid will increase the serum half-life of Cefuroxime. Cefuroxime does not interfere in enzyme based tests for glycosuria. Slight interference with copper reduction methods (Benedict's, Fehling's, Clinitest) may be observed. However, this should not lead to false-positive results, as may be experienced with some other cephalosporin. Cefuroxime may cause false-negative results, in the ferricyanide test for glucose. The antibiotic does not interfere the alkaline picrate assay for creatinine.

OVERDOSAGE AND TREATMENT :

Overdosage of cephalosporins can cause cerebral irritation leading to convulsions. Serum levels of Cefuroxime can be reduced by haemodialysis and peritoneal dialysis.

STORAGE:

Keep the container well closed. Store below 30°C. Protect from light. Freshly reconstituted solution is recommended.

**KEEP OUT OF REACH OF CHILDREN
JAUHI DARI KANAK-KANAK**

PACK QUANTITIES:

Single vial per box and 10 vials per box for 750mg & 1.5g.



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